CAUSAL DISCOVERY FROM COMPLEX SURVEY DATA: A SURVEY-WEIGHTED ORDINAL BAYESIAN NETWORK FOR SOCIAL DETERMINANTS OF HEALTH

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Social determinants of health (SDOH) are highly intercorrelated, making it difficult to differentiate root causes from mediating factors using standard associational methods. While causal discovery algorithms offer a data-driven approach to learn such causal structures, existing methods fail when applied to survey data. They rely on the assumption of independent and identically distributed data, whereas modern surveys employ unequal selection probabilities. To overcome this limitation for ordinal survey data, we introduce the survey-weighted ordinal Bayesian network, or `swa-oBn`. By integrating normalized Horvitz-Thompson weights and Kish's effective sample size into a Bayesian framework and exploiting a functional asymmetry between ordinal causes and effects, `swa-oBn` uniquely identifies causal directionality within Markov equivalence classes. We prove that the framework consistently recovers the true population-level causal graph in large samples, even under complex survey designs. We apply `swa-oBn` to national Veterans Affairs (VA) CAHPS survey data to map the causal architecture of Veteran SDOH and health outcomes. We demonstrate that ignoring the complex survey design misrepresents discrimination as a root cause of material deprivation and health outcomes. By explicitly modeling the sampling design, `swa-oBn` recovers the true structural hierarchy, revealing that unmet basic needs are the actual upstream hub, with discrimination acting solely as a downstream mediator to poor mental health. Complementary simulations confirm that ignoring complex survey designs produces severely biased causal graphs.

1. Introduction.

Social determinants of health (SDOH) such as financial instability, housing insecurity, and social isolation are highly intercorrelated in population surveys, creating a fundamental challenge for intervention design. Standard regression analyses document associations between unmet social needs and poor health outcomes (Braveman, Egerter and Williams, 2011; Duan-Porter et al., 2018), but they cannot differentiate upstream root causes from downstream mediating pathways (Glass et al., 2013; Athey, 2017). This distinction is critical for policy: if unmet basic needs are a root cause that cascades into discrimination, isolation, and poor mental health, then interventions targeting basic needs will have broad downstream benefits; if the causal ordering is different, so is the optimal allocation of resources.

Causal discovery algorithms offer a data-driven methodology for inferring directed causal structure from observational data (Spirtes, Glymour and Scheines, 2000; Pearl, 2009; Glymour, Zhang and Spirtes, 2019). Unlike structural equation modeling (Wright, 1921), which

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requires the causal architecture to be pre-specified, these algorithms learn the directed acyclic graph (DAG) from the data. However, applying causal discovery to survey data introduces a methodological problem that, to our knowledge, has not been addressed in the literature: complex surveys produce non-i.i.d. data, and ignoring the survey design yields distorted or biased underlying causal relationships.

To illustrate this problem concretely, consider a simple population causal chain $X \to Y \to Z$, where X, Y, and Z are ordinal variables representing basic financial need, access to resources, and well-being. In the population, X and Z are conditionally independent given Y. Now suppose a survey over-samples individuals with both high X (severe need) and low Z (poor well-being). The resulting sample over-represents the joint tail of (X, Z), inducing a spurious marginal dependence between X and Z that persists even after conditioning on Y. An unweighted causal discovery algorithm, confronted with this distorted distribution, will infer a direct edge $X \to Z$ (or $Z \to X$) that does not exist in the population, producing a causal graph that reflects the sampling mechanism rather than the data-generating process This bias is systematic, not random: it worsens with stronger oversampling and does not vanish as the sample size increases.

The problem is compounded by a second challenge specific to categorical data. Standard multinomial Bayesian networks (BNs) are non-identifiable: multiple DAGs that encode the same conditional independence relationships produce identical likelihoods (i.e., Markov equivalence), so the direction of many edges cannot be determined even with infinite data (Heckerman, Meek and Cooper, 1997; Chickering, 2002). Ni and Mallick (2022) resolved this identifiability problem for ordinal data by introducing the ordinal Bayesian network (`oBN`). They proved that the ordinal parameterization generically breaks Markov equivalence when all variables have three or more levels, enabling full DAG identifiability from observational data. Ni, Chen and Wang (2025) extended this to a bootstrapped multivariate algorithm with a BIC-based score and demonstrated its utility in a study of obsessive-compulsive disorder and depression. To our knowledge no other causal discovery algorithm achieves full DAG identifiability for a network of ordinal data. However, the `oBN` framework assumes i.i.d. data, does not support exogenous covariate adjustment, and uses a single BIC score that introduces a finite-sample orientation bias when ordinal variables have heterogeneous numbers of levels.

This paper addresses these gaps by introducing the *survey-weighted, covariate-adjusted ordinal Bayesian network*, or `swa-oBn`. The method makes four contributions over the existing `oBN` framework:

1. *Survey weighting.* We integrate normalized Horvitz–Thompson weights (Horvitz and Thompson, 1952) and Kish's effective sample size (Kish, 1965) into a pseudo-BIC scoring framework, enabling design-consistent structure learning from complex survey data.
2. *Two-step scoring.* We develop a scoring procedure that separates skeleton discovery (penalized pseudo-BIC) from edge orientation (unpenalized pseudo-likelihood). While the single-BIC approach of Ni and Mallick (2022); Ni, Chen and Wang (2025) is asymptotically valid regardless of whether ordinal scales are heterogeneous, the two-step procedure eliminates a systematic finite-sample bias that arises when the penalty counts differ across Markov-equivalent DAGs.
3. *Covariate adjustment.* Measured demographic confounders are absorbed into every nodewise regression without expanding the DAG, reducing confounding bias while preserving the ordinal identifiability guarantee.
4. *Structural constraints.* Domain knowledge is encoded via blacklisted and whitelisted edges, restricting the search space to scientifically plausible graphs.

We prove that the `swa-oBn` consistently recovers the true population-level DAG in large samples under a joint superpopulation-and-design asymptotic framework, establishing skeleton consistency via the penalized pseudo-BIC and orientation consistency via the unpenalized pseudo-likelihood.

We apply the `swa-oBn` to national data from the Veterans Health Administration (VHA) Consumer Assessment of Healthcare Providers and Systems (CAHPS) survey to map the causal architecture of social determinants of health among U.S. Veterans. Recent analyses of this dataset have documented substantial variation in social need prevalence (Frank et al., 2025) across age and race-ethnicity-sex, and shown that standard social risk screening under-detects needs among certain demographic subgroups (Russell et al., 2026). These analyses rely on associational methods that assume rather than discover causal structure. The `swa-oBn` addresses this gap, revealing that unmet basic needs act as an upstream causal hub, while discrimination and isolation serve as the proximal mediating pathway to poor mental health and well-being.

The remainder of this paper is organized as follows. Section 2 introduces the CAHPS data and the scientific questions motivating the analysis. Section 3 reviews ordinal Bayesian networks. Section 4 develops the `swa-oBn` methodology. Section 5 establishes theoretical properties. Section 6 presents simulation studies. Section 7 reports the application to VHA data. Section 8 concludes with a discussion.

2. Data and Scientific Questions.

2.1. *The Veterans Health Administration and social determinants of health.* The Veterans Health Administration (VHA) is the largest integrated health care system in the United States, providing care to over nine million enrolled Veterans annually (U.S. Department of Veterans Affairs, 2026). Understanding and addressing social determinants of health (SDOH) among this population has become a major policy objective (Daniel, Bornstein and Kane, 2021). The existing literature on SDOH in the Veteran population has extensively documented that unmet social needs are associated with poor health outcomes (Braveman, Egerter and Williams, 2011; Duan-Porter et al., 2018). Recent analyses of the VHA patient population have revealed substantial variation in social need prevalence: compared with White male Veterans, Black male Veterans report elevated needs in 11 of 13 domains, and need for support managing discrimination is elevated across all non-White-male subgroups (Frank et al., 2025; Lamba et al., 2025). Furthermore, standard social risk screening instruments under-detect need for support among certain demographic subgroups, particularly Black male Veterans (Russell et al., 2026). These findings confirm that SDOH are pervasive and heterogeneous, but the analyses underlying them are entirely associational. Regression models report which needs are correlated with which outcomes; they do not reveal the causal ordering among needs, leaving policymakers without the directional knowledge required to prioritize interventions effectively. This paper addresses the following policy questions that standard associational methods cannot answer:

1. Which unmet social needs are upstream root causes of other needs, and which are downstream consequences?
2. Among the needs causally linked to poor health, which function as *distal* causes (i.e., operating on outcomes indirectly through mediating needs) and which function as *proximal* causes that transmit upstream effects directly to health?

The distal–proximal distinction is not merely descriptive; it is a property of the causal topology that only a DAG-based analysis can reveal. A need that is strongly correlated with mental health may be causally distal (acting through intermediate mediators, so that interventions bypass most of its effect unless the mediators are also addressed) or causally proximal (directly adjacent to the outcome, so that targeting it yields immediate benefit). Associational methods cannot distinguish these cases; the `swa-oBn` can.

2.2. *Survey design and data.* We utilize data from the Consumer Assessment of Healthcare Providers and Systems (CAHPS) Clinician and Group Visit Survey, a standardized national instrument developed by the Agency for Healthcare Research and Quality (AHRQ) and administered by the VHA as part of its Survey of Healthcare Experiences of Patients (SHEP) program (Dyer et al., 2012; **?**; Hausmann et al., 2023). The survey was fielded to a stratified random sample of VHA primary care patients seen in January or February 2023. To ensure adequate statistical power for intersectional comparisons, the design deliberately oversampled women and Black and Hispanic Veterans, producing six race-ethnicity-sex strata with approximately 900 respondents each (Frank et al., 2025). The analytic sample comprises $n = 5{,}396$ respondents (complete-case analysis). Survey weights account for the sampling design and survey nonresponse and vary substantially: the ratio of the largest to smallest weight is approximately 990:1, with a coefficient of variation of 1.70. Kish's effective sample size is $n_{\text{eff}} = 1{,}387$, yielding a design effect of 3.89. This effective sample size governs the penalty term in the pseudo-BIC (Section 4.2) and reflects a nearly fourfold reduction in effective information relative to a simple random sample of the same size.

2.3. *Variable construction.* The CAHPS survey assesses need for support across 13 social domains using a three-level ordinal scale: "no support needed" (level 1), "needed support and got it" (level 2), and "needed support but did not get it" (level 3). Higher values represent more severe unmet need. The companion analyses of Frank et al. (2025) and Russell et al. (2026) collapsed levels 2 and 3 into a binary indicator of need for the purpose of estimating prevalence ratios. The present analysis retains the full ordinal scale, which is essential: the identifiability result of Ni and Mallick (2022) that underlies our framework requires variables with three or more levels, and the three-level response captures the substantively important distinction between met and unmet need.

A fundamental challenge in applying causal discovery to survey instruments is the high dimensionality and severe collinearity of item-level data. To construct conceptually distinct nodes for the Bayesian network, we analyzed the item-level polychoric correlation structure. Items exhibiting polychoric correlations $\rho \geq 0.75$ were aggregated into composite ordinal domains by taking the most severe level of need across the constituent items, thus preserving the original three-level ordinal scale. This yielded three composite nodes: *basic needs* (food, housing, paying for basics; $\rho = 0.85$–0.95), *work/education* (finding work, additional education; $\rho = 0.82$), and *discrimination/isolation* (discrimination, social isolation, loneliness; $\rho = 0.75$–0.96). Five items falling below this threshold were retained as standalone nodes: *caregiving*, *childcare*, *transportation*, *internet access*, and *legal assistance*. The downstream health outcomes were captured by two additional ordinal nodes: *well-being* (four levels) and *mental health* (five levels). The complete node set consists of $q = 10$ ordinal variables with 3–5 levels each. The polychoric correlation matrix and clustering are presented in the Supplementary Material.

REMARK 1. *The polychoric correlation threshold $\rho \geq 0.75$ was chosen in collaboration with domain experts to balance construct validity with network interpretability. We treat this as a deliberate methodological trade-off, accepting an approximate working model in exchange for an interpretable graph. The bootstrap procedure of Section 4.4 provides robustness against orientation artifacts that might arise from this approximation. A fuller discussion of the implications of this choice appears in Section 8.*

2.4. *Covariate adjustment and structural constraints.* Observational causal discovery is subject to the assumption of causal sufficiency: all common causes of any pair of variables in the DAG must be measured. To mitigate confounding by demographic characteristics, our analysis adjusts for age (categorized into four bands) and a combined race-ethnicity-sex indicator, the same covariates used in the companion analyses of this dataset (Frank et al., 2025; Lamba et al., 2025; Russell et al., 2026). These variables are not treated as nodes in the DAG; instead, they form an exogenous covariate matrix $\boldsymbol{Z}$ that is conditioned on in every node-wise regression (Section 4.3), ensuring the discovered edges represent relationships adjusted for demographic confounding. A discussion of what $\boldsymbol{Z}$ does and does not absorb is deferred to Section 8.

Finally, we integrate domain knowledge via a structural blacklist. All directed edges originating from the two outcome nodes (well-being and mental health) and pointing toward any of the eight social need nodes are forbidden. This encodes the minimal substantive assumption that structural social needs are upstream of health states. All relationships among the social need variables themselves are left entirely unconstrained.

3. Background: Ordinal Bayesian Networks. A Bayesian network (`Bn`) represents the joint distribution of random variables $\boldsymbol{X} = (X_1, \ldots, X_q)$ through a directed acyclic graph (DAG) $\mathcal{G} = (V, E)$ and an associated factorization $P(\boldsymbol{X} \mid \mathcal{G}) = \prod_{j=1}^{q} P(X_j \mid \boldsymbol{X}_{\mathrm{pa}(j)})$, where $\mathrm{pa}(j)$ denotes the parent set of node j in $\mathcal{G}$ (Pearl, 2009). Under the standard assumptions of causal sufficiency and faithfulness — stated formally in Section 5.1 — the edges of $\mathcal{G}$ admit a causal interpretation and the graph can in principle be recovered from observational data. For categorical variables, each conditional $P(X_j \mid \boldsymbol{X}_{\mathrm{pa}(j)})$ is typically specified by a multinomial distribution. A well-known limitation of this parameterization is that it is identifiable only up to a Markov equivalence class: all DAGs encoding the same conditional independence relations produce the same likelihood, leaving the direction of many edges undetermined (Heckerman, Meek and Cooper, 1997; Chickering, 2002).

When the categorical variables are ordinal, the multinomial parameterization discards informative structure. Ni and Mallick (2022) proposed the ordinal Bayesian network (`oBn`), which replaces multinomial conditionals with cumulative link regressions. For an ordinal variable X_j with L_j levels and parent set $\mathrm{pa}(j)$, the `oBn` specifies

$$\Pr(X_j \leq \ell \mid \boldsymbol{X}_{\mathrm{pa}(j)}) = \Phi\left(\gamma_{j\ell} - \sum_{k \in \mathrm{pa}(j)} \beta_{jk} X_k - \alpha_j\right), \quad \ell = 1, \ldots, L_j, \tag{1}$$

where Φ is the standard normal cumulative distribution function (any strictly increasing link F may be used; we adopt Φ for concreteness and consistency with Ni and Mallick (2022); Ni, Chen and Wang (2025)), $\gamma_{j1}, \ldots, \gamma_{j,L_j-1}$ are ordered cutpoints with $\gamma_{j1} = 0$ for identifiability, α_j is an intercept, and β_{jk} captures the effect of parent X_k on X_j. When a parent X_k enters as a predictor with L_k levels, separate coefficients $\beta_{jk1}, \ldots, \beta_{jk,L_k-1}$ replace the scalar β_{jk}, with $\beta_{jk1} = 0$ as a reference constraint. The total number of free parameters under DAG $\mathcal{G}$ is

$$K_{\mathcal{G}} = \sum_{j=1}^{q} \left(L_j - 1 + \sum_{k \in \mathrm{pa}(j)} (L_k - 1)\right). \tag{2}$$

This count plays a role in the BIC penalty and will be important for motivating the two-step scoring procedure of Section 4.1.

The central property of the `oBn` is generic identifiability. Ni and Mallick (2022) proved that if $X_j \to X_k$ is the true data-generating direction and both variables have three or more levels, then for Lebesgue-almost all parameter values, the reverse model $X_k \to X_j$ cannot reproduce the same joint distribution under the ordinal parameterization. That is, the sets of joint distributions reachable under $X_j \to X_k$ and $X_k \to X_j$ are generically distinct. Ni, Chen and Wang (2025) extended this to the multivariate setting and introduced a BIC-based greedy hill-climbing algorithm with bootstrap uncertainty quantification. This identifiability result is what makes score-based structure learning viable for ordinal data: the true DAG can be distinguished from all Markov-equivalent alternatives, not merely located within an equivalence class. However, the framework of Ni and Mallick (2022); Ni, Chen and Wang (2025) assumes i.i.d. data, does not accommodate survey weights or exogenous covariates, and uses a single BIC score for all search moves. The next section develops the extensions required to address these limitations.

4. Survey-Weighted Ordinal Causal Discovery. We now develop the survey-weighted, covariate-adjusted ordinal Bayesian network, or `swa-oBn`, which extends the `oBn` framework of Ni and Mallick (2022); Ni, Chen and Wang (2025) to the setting of complex survey data with measured confounders and domain knowledge. The method comprises four components: a two-step scoring procedure that separates skeleton discovery from edge orientation (Section 4.1), survey weighting via a pseudo-BIC with design-effect-corrected penalty (Section 4.2), covariate adjustment through an additive regression term (Section 4.3), and bootstrap uncertainty quantification (Section 4.4). Structural constraints are incorporated as a remark within Section 4.1. The full algorithm is summarized as pseudocode in the Supplementary Material.

4.1. *Two-step scoring procedure.* Let $\boldsymbol{x} = (\boldsymbol{x}_1, \dots, \boldsymbol{x}_n)$ denote n observed realizations of $\boldsymbol{X}$. Under the `oBn`factorization with cumulative probit conditionals (1), the log-likelihood decomposes as a sum of node-wise contributions,

$$\ell(\boldsymbol{\theta}; \mathcal{G}) = \sum_{j=1}^{q} \ell_j(\boldsymbol{\theta}_j; \mathrm{pa}(j)), \quad \text{where} \quad \ell_j(\boldsymbol{\theta}_j; \mathrm{pa}(j)) = \sum_{i=1}^{n} \log p(x_{ij} \mid \boldsymbol{x}_{i,\mathrm{pa}(j)}, \boldsymbol{\theta}_j), \tag{3}$$

and $\boldsymbol{\theta}_j = (\boldsymbol{\gamma}_j, \boldsymbol{\beta}_j, \alpha_j)$ collects the cutpoints, regression coefficients, and intercept for node j. This decomposition is central to score-based structure learning: candidate DAGs can be compared by evaluating only the node-wise scores that change under a proposed edge modification. Ni and Mallick (2022); Ni, Chen and Wang (2025) scored candidate graphs using the Bayesian information criterion (BIC),

$$\mathrm{BIC}(\mathcal{G}) = -2\,\ell(\hat{\boldsymbol{\theta}}; \mathcal{G}) + K_{\mathcal{G}} \log n, \tag{4}$$

where $K_{\mathcal{G}}$ is given by (2) and $\hat{\boldsymbol{\theta}}$ denotes the maximum likelihood estimate. When all ordinal variables share a common number of levels L, the parameter count for the regression of X_j on parent X_k is the same regardless of whether the edge is oriented as $k \to j$ or $j \to k$, so the BIC penalties cancel within a Markov equivalence class and orientation is governed entirely by the likelihood. When ordinal scales are heterogeneous, however, the regression of X_j on X_k contributes $L_k - 1$ slope parameters while the reverse contributes $L_j - 1$. Two Markov-equivalent DAGs that differ only in the direction of a single edge then receive different BIC penalties, introducing a systematic finite-sample bias into orientation decisions. We note that this bias is asymptotically negligible — the penalty discrepancy is $O(\log n)$ while the likelihood ratio is $O(n)$ — so the single-BIC approach of Ni and Mallick (2022); Ni, Chen and Wang (2025) remains asymptotically valid. Nevertheless, the two-step procedure we describe next eliminates this bias entirely and yields a clean separation of concerns: the penalty controls sparsity, and the likelihood controls orientation.

We separate the scoring criterion across two types of search moves:

- *Addition and deletion* moves alter the skeleton of the graph. These are evaluated using the full penalized BIC (4) (or its survey-weighted analog (7) below), so that the complexity penalty controls the number of edges and encourages parsimony.

- *Reversal* moves change only the direction of an existing edge. These are evaluated using the unpenalized quantity $-2\,\ell(\hat{\boldsymbol{\theta}};\mathcal{G})$, reducing the comparison between $k \to j$ and $j \to k$ to a pure likelihood ratio. Because the ordinal parameterization is generically identifiable (Ni and Mallick, 2022), the correctly oriented edge achieves a strictly higher likelihood for almost all data-generating parameter values, regardless of the number of levels.

The search proceeds by greedy hill-climbing (Chickering, 2002; Scutari, 2010). Starting from an initial DAG, each iteration evaluates all candidate single-edge modifications — additions, deletions, and reversals — subject to the constraint that the resulting graph remains acyclic. The modification yielding the largest score improvement is applied, and the process continues until no improving move exists. Because greedy hill-climbing is guaranteed to find only a local optimum, we run the search from R random initializations and select the solution with the lowest penalized score across all restarts.

REMARK 2 (Computational efficiency). *The per-iteration cost of the greedy search is dominated by the ordinal regressions required to evaluate candidate moves. We reduce this cost through hash-map caching of node-wise scores: each unique combination of node, parent set, and scoring mode (penalized or unpenalized) is computed at most once and retrieved from cache in subsequent iterations. Combined with the node-wise decomposition* (3)*, which limits recomputation to the nodes whose parent sets change, this yields substantial speedups over the implementation of Ni and Mallick (2022); Ni, Chen and Wang (2025).*

REMARK 3 (Structural constraints). *Domain knowledge can be encoded through a* blacklist $\mathcal{B}$ *of forbidden directed edges and a* whitelist $\mathcal{W}$ *of required directed edges. These constraints are enforced at each iteration of the hill-climbing search: blacklisted edges are never proposed for addition and reversals that would create a blacklisted edge are skipped; whitelisted edges are included in the initial graph and cannot be deleted or reversed.*

4.2. *Survey weighting.* When inclusion probabilities are associated with the variables under study, the sampling mechanism is *informative* (Pfeffermann, 1993) and parameter estimates obtained by treating the sample as i.i.d. can be inconsistent for the corresponding population quantities. In the context of causal discovery, this means that the graph learned from an unweighted analysis may reflect the sampling design rather than the population-generating causal structure.

The standard correction is pseudo-maximum likelihood estimation (Binder, 1983; Skinner, 1989). Let S denote a probability sample of size n drawn from a finite population of size N, and let $\pi_i = \Pr(i \in S)$ denote the first-order inclusion probability for unit i, with survey weight $w_i = 1/\pi_i$. We replace the log-likelihood (3) with the pseudo-log-likelihood

$$\ell^w(\boldsymbol{\theta};\mathcal{G}) = \sum_{i\in S} w_i \log p(\boldsymbol{x}_i \mid \mathcal{G},\boldsymbol{\theta}), \tag{5}$$

which, by the Horvitz–Thompson property, is design-unbiased for the census log-likelihood $\sum_{i=1}^{N} \log p(\boldsymbol{x}_i \mid \mathcal{G},\boldsymbol{\theta})$. Because the oBnfactorizes node-wise, the pseudo-log-likelihood inherits the same decomposition:

$$\ell^w(\boldsymbol{\theta};\mathcal{G}) = \sum_{j=1}^{q} \ell^w_j(\boldsymbol{\theta}_j;\mathrm{pa}(j)), \quad \text{where} \quad \ell^w_j(\boldsymbol{\theta}_j;\mathrm{pa}(j)) = \sum_{i\in S} w_i \log p(x_{ij} \mid \boldsymbol{x}_{i,\mathrm{pa}(j)},\boldsymbol{\theta}_j). \tag{6}$$

The pseudo-BIC for graph $\mathcal{G}$ is

$$\mathrm{pBIC}(\mathcal{G}) = -2\,\ell^w(\hat{\boldsymbol{\theta}}^w;\mathcal{G})\,\frac{n_{\mathrm{eff}}}{n} \;+\; K_{\mathcal{G}} \log(n_{\mathrm{eff}}), \tag{7}$$

where $\hat{\boldsymbol{\theta}}^w$ is the pseudo-maximum likelihood estimate and n_{eff} is Kish's effective sample size (Kish, 1965),

$$n_{\text{eff}} = \frac{\left(\sum_{i\in S} w_i\right)^2}{\sum_{i\in S} w_i^2}. \tag{8}$$

The factor n_{eff}/n in the likelihood term is a design-effect correction: when the weights are internally normalized to sum to n, the raw pseudo-log-likelihood is on the sample scale, and rescaling by n_{eff}/n brings it to the effective-information scale so that the likelihood and penalty are commensurate (Lumley and Scott, 2015). When all weights are equal, $n_{\text{eff}} = n$ and the pseudo-BIC reduces to the standard BIC. Both steps of the two-step procedure use survey-weighted quantities: Step 1 (add/delete) evaluates the full pseudo-BIC (7), while Step 2 (reversal) evaluates only the rescaled pseudo-log-likelihood $-2\,\ell^w(\hat{\boldsymbol{\theta}}^w;\mathcal{G})\,(n_{\text{eff}}/n)$, omitting the penalty.

REMARK 4 (Relation to the theoretical pseudo-BIC). *The pseudo-BIC* (7) *uses normalized weights (*$\sum_{i\in S} w_i = n$*) with likelihood rescaling* n_{eff}/n *and penalty* $\log(n_{\text{eff}})$*; Section* 5 *analyzes the equivalent formulation using raw weights (*$w_i = 1/\pi_i$*) with penalty* ρ_n*. Under normalized weights the likelihood term is on the* n_{eff} *scale; under raw weights it is on the* N *scale. The ratio* N/n_{eff} *is graph-invariant, so the two formulations yield the same minimizer* $\arg\min_{\mathcal{G}} \mathrm{pBIC}(\mathcal{G})$*. The normalized-weight formulation is numerically preferable because both likelihood and penalty are on the same effective-information scale.*

4.3. *Covariate adjustment.* Measured covariates $\boldsymbol{Z} = (Z_1,\ldots,Z_r) \in \mathbb{R}^r$ may confound the relationships among the ordinal variables $\boldsymbol{X}$ but are not themselves objects of causal discovery. We absorb their influence by extending the node-wise ordinal regression (1) to include $\boldsymbol{Z}$ as an additive predictor in every node:

$$\Pr(X_j \le \ell \mid \boldsymbol{X}_{\mathrm{pa}(j)}, \boldsymbol{Z}) = \Phi\left(\gamma_{j\ell} - \sum_{k\in \mathrm{pa}(j)} \beta_{jk} X_k - \boldsymbol{Z}'\boldsymbol{\delta}_j - \alpha_j\right), \quad \ell = 1,\ldots,L_j, \tag{9}$$

where $\boldsymbol{\delta}_j$ is a node-specific coefficient vector. Because $\boldsymbol{Z}$ enters every node regression with its own $\boldsymbol{\delta}_j$, the covariates are allowed to affect each variable differently. Crucially, $\boldsymbol{Z}$ does not appear as a node in $\mathcal{G}$: the search algorithm adds, deletes, and reverses edges only among the q variables in $\boldsymbol{X}$, so the DAG search space remains $O(q^2)$ candidate edges rather than $O((q+r)^2)$. The covariate-adjusted pseudo-log-likelihood for node j is

$$\ell_j^w(\boldsymbol{\theta}_j, \boldsymbol{\delta}_j; \mathrm{pa}(j)) = \sum_{i\in S} w_i \log p(x_{ij} \mid \boldsymbol{x}_{i,\mathrm{pa}(j)}, \boldsymbol{z}_i, \boldsymbol{\theta}_j, \boldsymbol{\delta}_j), \tag{10}$$

where $p(\cdot)$ is evaluated under (9). The pseudo-BIC and the two-step scoring procedure of Sections 4.1–4.2 apply unchanged, with $\boldsymbol{\delta}_j$ absorbed into the parameter vector. This extension requires that the causal sufficiency assumption be strengthened: all common causes of any pair (X_j, X_k) must lie in $\boldsymbol{X} \cup \boldsymbol{Z}$ rather than $\boldsymbol{X}$ alone. We show in Proposition 1 (Section 5) that generic identifiability is preserved under this extension: for each fixed value of $\boldsymbol{Z}$, the covariate-adjusted model (9) reduces to an `oBn`with shifted intercepts, so the identifiability result of Ni and Mallick (2022) continues to hold.

4.4. *Bootstrap uncertainty quantification.* A single run of the hill-climbing algorithm produces a point-estimate DAG $\hat{\mathcal{G}}$. To quantify uncertainty in the learned structure, we adopt the nonparametric bootstrap (Efron and Tibshirani, 1993). For each replicate $b = 1,\ldots,B$, we resample n observation-weight-covariate tuples $(\boldsymbol{x}_i, w_i, \boldsymbol{z}_i)$ with replacement, recompute n_{eff} from the resampled weights, and re-run the full two-step search to obtain $\hat{\mathcal{G}}^{(b)}$. The edge inclusion probability for the directed edge $j \to k$ is

$$\hat{p}_{jk} = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\left(j \to k \in \hat{\mathcal{G}}^{(b)}\right), \tag{11}$$

and a consensus graph is constructed by retaining edges with $\hat{p}_{jk}$ exceeding a chosen threshold. This procedure follows Ni, Chen and Wang (2025), adapted to the survey-weighted setting. By the consistency result of Section 5, each sufficiently large resample recovers the true DAG $\mathcal{G}^*$, so the edge inclusion probabilities converge to zero or one for all edges as $n \to \infty$. See supplement for the complete `swa-oBn` algorithm.

5. Theoretical Properties. We establish the large-sample properties of the `swa-oBn` under a joint asymptotic framework that accounts for two sources of randomness: the superpopulation ξ generating the finite population, and the sampling design selecting units from it. Let $U_N = \{1, \dots, N\}$ denote a finite population of size N, generated as N independent draws from ξ, with true causal DAG $\mathcal{G}^*$ and parameters $\boldsymbol{\theta}^*$. From U_N, a probability sample S_n of size $n = n(N)$ is drawn according to a sampling design $p_N(S)$ with known first-order inclusion probabilities $\pi_i = \Pr(i \in S_n)$ and survey weights $w_i = 1/\pi_i$. We require $N \to \infty$ and $n \to \infty$ with sampling fraction $n/N \to f \in [0, 1)$. All convergence statements are under the joint $(\xi \times p_N)$-probability. Proofs are given in the Supplementary Material.

We use $\boldsymbol{\theta}^*_{\mathcal{G}}$ to denote the superpopulation maximizer of $Q_\xi(\cdot \mid \mathcal{G})$, i.e., $\boldsymbol{\theta}^*_{\mathcal{G}} = \arg\max_{\boldsymbol{\theta}} Q_\xi(\boldsymbol{\theta} \mid \mathcal{G})$; when $\mathcal{G} = \mathcal{G}^*$, this coincides with the true data-generating parameter, which we denote simply by $\boldsymbol{\theta}^*$. For a misspecified DAG $\mathcal{G}' \neq \mathcal{G}^*$, we use $\boldsymbol{\theta}^\dagger_{\mathcal{G}'} \equiv \boldsymbol{\theta}^*_{\mathcal{G}'}$ to emphasize its role as the KL-projection rather than a truth-level parameter. The *conditional* (covariate-pointwise) KL-projection is denoted $\boldsymbol{\theta}^\dagger(\boldsymbol{z}) = \arg\max_{\boldsymbol{\theta}} \sum_{\boldsymbol{x}} P(\boldsymbol{x} \mid \mathcal{G}^*, \boldsymbol{\theta}^*, \boldsymbol{z}) \log P(\boldsymbol{x} \mid \mathcal{G}', \boldsymbol{\theta}, \boldsymbol{z})$, which in general differs from the unconditional $\boldsymbol{\theta}^\dagger_{\mathcal{G}'}$.

5.1. *Assumptions.* We organize the assumptions into four groups: causal assumptions (1–3), parametric assumptions (4–6), survey design assumptions (7–9), and a regularity assumption (10).

ASSUMPTION 1 (Causal Sufficiency). *All common causes of any pair (X_j, X_k) in $\boldsymbol{X} = (X_1, \dots, X_q)$ are included in the measured variables $\boldsymbol{X} \cup \boldsymbol{Z}$.*

ASSUMPTION 2 (Causal Markov Property). *The superpopulation distribution $P_\xi(\boldsymbol{X})$ factorizes with respect to the true causal DAG $\mathcal{G}^*$: $P_\xi(\boldsymbol{X}) = \prod_{j=1}^{q} P_\xi(X_j \mid \boldsymbol{X}_{\mathrm{pa}(j)})$.*

ASSUMPTION 3 (Faithfulness). *Every conditional independence in P_ξ is entailed by the Markov property of $\mathcal{G}^*$: $X_A \perp\!\!\!\perp X_B \mid X_C$ in P_ξ if and only if A and B are d-separated by C in $\mathcal{G}^*$.*

ASSUMPTION 4 (Ordinal Regression Specification). *Each conditional distribution under the true DAG $\mathcal{G}^*$ follows a cumulative link model:*

$$P_\xi(X_j \le \ell \mid \boldsymbol{X}_{\mathrm{pa}(j)}, \boldsymbol{Z}) = F\left(\gamma_{j\ell} - \sum_{k \in \mathrm{pa}(j)} \beta_{jk} X_k - \boldsymbol{\delta}_j' \boldsymbol{Z} - \alpha_j\right), \quad \ell = 1, \dots, L_j,$$

where F is a known, strictly increasing link function, $\gamma_{j1} < \cdots < \gamma_{j,L_j} = \infty$ are threshold parameters with $\gamma_{j1} = 0$, β_{jk} are parent effect parameters with $\beta_{jk,1} = 0$, $\boldsymbol{\delta}_j$ are covariate adjustment parameters, and α_j is an intercept. We require $L_j \ge 3$ for all $j = 1, \dots, q$.

REMARK 5. *The restriction $L_j \geq 3$ is necessary: if $L_j = 2$, the cumulative link model collapses to a standard binary regression and causal directionality is identifiable only up to the Markov equivalence class (Ni and Mallick, 2022).*

ASSUMPTION 5 (Ordinal Identifiability). *Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be any two Markov-equivalent DAGs on $\boldsymbol{X}$, and let $\boldsymbol{\theta}_1$ denote a parameter under $\mathcal{G}_1$ in the no-covariate oBN (Assumption 4 with $\boldsymbol{\delta}_j = \mathbf{0}$ for all j). Define the pseudo-true parameter $\boldsymbol{\theta}_2^\dagger(\boldsymbol{\theta}_1) = \arg\max_{\boldsymbol{\theta}} \sum_{\boldsymbol{x}} P(\boldsymbol{x} \mid \mathcal{G}_1, \boldsymbol{\theta}_1) \log P(\boldsymbol{x} \mid \mathcal{G}_2, \boldsymbol{\theta})$. Then the set of $\boldsymbol{\theta}_1$ for which $P(\boldsymbol{X} \mid \mathcal{G}_1, \boldsymbol{\theta}_1) \equiv P(\boldsymbol{X} \mid \mathcal{G}_2, \boldsymbol{\theta}_2^\dagger(\boldsymbol{\theta}_1))$ is a proper algebraic subvariety of $\Theta_{\mathcal{G}_1}$ of Lebesgue measure zero. We assume the true no-covariate parameter $\boldsymbol{\theta}^*$ lies outside this rare set.*

Assumption 5 is the central identifiability result of Ni and Mallick (2022) for the bivariate oBN; its extension to multivariate DAGs follows from the pairwise decomposition argument of Ni, Chen and Wang (2025). Proposition 1 lifts this generic identifiability to the covariate-adjusted model.

ASSUMPTION 6 (Non-Degeneracy). *$P_\xi(X_j = \ell \mid \boldsymbol{X}_{\mathrm{pa}(j)} = \boldsymbol{x}_{\mathrm{pa}(j)}, \boldsymbol{Z} = \boldsymbol{z}) > 0$ for all j, all levels ℓ, all parent configurations $\boldsymbol{x}_{\mathrm{pa}(j)}$, and all $\boldsymbol{z}$ in the support of $\boldsymbol{Z}$.*

ASSUMPTION 7 (Positivity). *Every unit $i \in U_N$ has a known, strictly positive first-order inclusion probability: $\pi_i = \Pr(i \in S_n) > 0$.*

ASSUMPTION 8 (Bounded Weights). *There exist constants $0 < c \leq C < \infty$, independent of N, such that $c \leq n w_i / N \leq C$ for all $i \in U_N$, ensuring that no single unit dominates the pseudo-likelihood.*

ASSUMPTION 9 (Design Consistency). *For any bounded, measurable function g on the joint sample space of $(\boldsymbol{X}, \boldsymbol{Z})$,*

$$\frac{1}{N} \sum_{i \in S_n} w_i \, g(\boldsymbol{x}_i, \boldsymbol{z}_i) - \frac{1}{N} \sum_{i \in U_N} g(\boldsymbol{x}_i, \boldsymbol{z}_i) \xrightarrow{p} 0$$

under the design distribution p_N, for each fixed realization of U_N.

Assumption 9 is satisfied by standard probability sampling designs under Assumptions 7–8; see Fuller (2009), Ch. 1, and Särndal, Swensson and Wretman (1992).

ASSUMPTION 10 (Regularity). *For each DAG $\mathcal{G}$: (i) the parameter space $\Theta_{\mathcal{G}}$ is compact; (ii) $\log P(\boldsymbol{x} \mid \mathcal{G}, \boldsymbol{\theta}, \boldsymbol{z})$ is twice continuously differentiable in $\boldsymbol{\theta}$ for each $(\boldsymbol{x}, \boldsymbol{z})$; (iii) the superpopulation expected log-likelihood $Q_\xi(\boldsymbol{\theta} \mid \mathcal{G}) = \mathbb{E}_\xi[\log P(\boldsymbol{X} \mid \mathcal{G}, \boldsymbol{\theta}, \boldsymbol{Z})]$ has a unique maximizer $\boldsymbol{\theta}_{\mathcal{G}}^*$ with nonsingular Hessian; and (iv) $\boldsymbol{\theta} \mapsto \log P(\boldsymbol{x} \mid \mathcal{G}, \boldsymbol{\theta}, \boldsymbol{z})$ satisfies a Lipschitz condition uniformly over $(\boldsymbol{x}, \boldsymbol{z})$.*

For the oBN with finitely many ordinal levels, condition (iv) is automatic since $\boldsymbol{X}$ has finite support and the link function F is smooth.

5.2. *Main Results.* Define the survey-weighted pseudo-log-likelihood under DAG $\mathcal{G}$ as

$$\tilde{\ell}(\boldsymbol{\theta} \mid \mathcal{G}) = \sum_{i \in S_n} w_i \log P(\boldsymbol{x}_i \mid \mathcal{G}, \boldsymbol{\theta}, \boldsymbol{z}_i), \tag{12}$$

the pseudo-MLE as $\hat{\boldsymbol{\theta}}^{w}_{\mathcal{G}} = \arg\max_{\boldsymbol{\theta}} \tilde{\ell}(\boldsymbol{\theta} \mid \mathcal{G})$, and the survey-weighted pseudo-BIC as

$$\widetilde{\mathrm{BIC}}(\mathcal{G}) = -2\,\tilde{\ell}(\hat{\boldsymbol{\theta}}^{w}_{\mathcal{G}} \mid \mathcal{G}) + K_{\mathcal{G}} \cdot \rho_n, \tag{13}$$

where $K_{\mathcal{G}} = \sum_{j=1}^{q}(L_j - 1 + \sum_{k\in \mathrm{pa}(j)}(L_k - 1) + \dim(\boldsymbol{\delta}_j))$ is the number of free parameters and ρ_n is a penalty sequence satisfying $\rho_n \to \infty$ and $\rho_n/N \to 0$. In practice, $\rho_n = \log(n_{\mathrm{eff}})$.

PROPOSITION 1 (Identifiability under covariate adjustment). *Suppose Assumption 5 holds for the no-covariate oBN. Let $\mathcal{G}^*$ and $\mathcal{G}'$ be Markov-equivalent DAGs with $\mathcal{G}' \neq \mathcal{G}^*$. Then the set of $(\boldsymbol{\theta}^*, \boldsymbol{\delta}^*) \in \Theta_{\mathcal{G}^*} \times \mathbb{R}^{rq}$ for which*

$$\mathbb{E}_{\boldsymbol{Z}}\Big[\mathrm{KL}\Big(P_{\xi}(\boldsymbol{X} \mid \mathcal{G}^*, \boldsymbol{\theta}^*, \boldsymbol{\delta}^*, \boldsymbol{Z}) \,\big\|\, P(\boldsymbol{X} \mid \mathcal{G}', \boldsymbol{\theta}^{\dagger}(\boldsymbol{Z}), \boldsymbol{Z})\Big)\Big] = 0$$

has Lebesgue measure zero. We assume throughout that the true parameters $(\boldsymbol{\theta}^*, \boldsymbol{\delta}^*)$ of the data-generating model lie outside this exceptional set, in which case the expected KL divergence is strictly positive.

PROPOSITION 2 (Pseudo-MLE Consistency). *Under Assumptions 4 and 6–10, for each DAG $\mathcal{G}$,*

$$\hat{\boldsymbol{\theta}}^{w}_{\mathcal{G}} \xrightarrow{p} \boldsymbol{\theta}^*_{\mathcal{G}}$$

under the joint $(\xi \times p_N)$-probability, where $\boldsymbol{\theta}^*_{\mathcal{G}} = \arg\max_{\boldsymbol{\theta}} Q_{\xi}(\boldsymbol{\theta} \mid \mathcal{G})$.

The two-step structure of this proof — design consistency followed by superpopulation convergence — is characteristic of the joint asymptotic framework for survey data (Fuller, 2009; Binder, 1983).

THEOREM 5.1 (Skeleton Consistency). *Under Assumptions 1–10, let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ any DAG not Markov-equivalent to $\mathcal{G}^*$. Then*

$$\Pr\Big(\widetilde{\mathrm{BIC}}(\mathcal{G}^*) < \widetilde{\mathrm{BIC}}(\mathcal{G}')\Big) \to 1.$$

Theorem 5.1 validates Step 1 of the `swa-oBn` algorithm: the penalized pseudo-BIC correctly recovers the skeleton of $\mathcal{G}^*$.

THEOREM 5.2 (Orientation Consistency). *Under Assumptions 1–10, let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ any DAG Markov-equivalent to $\mathcal{G}^*$ with $\mathcal{G}' \neq \mathcal{G}^*$. Then*

$$\Pr\Big(\tilde{\ell}(\hat{\boldsymbol{\theta}}^{w}_{\mathcal{G}^*} \mid \mathcal{G}^*) > \tilde{\ell}(\hat{\boldsymbol{\theta}}^{w}_{\mathcal{G}'} \mid \mathcal{G}')\Big) \to 1.$$

Theorem 5.2 validates Step 2: the unpenalized pseudo-likelihood correctly orients all edges within the Markov equivalence class. Because no penalty term appears in the orientation comparison, the result holds regardless of whether the ordinal variables have homogeneous or heterogeneous numbers of levels.

COROLLARY 1 (Full DAG Recovery). *Under Assumptions 1–10, the two-step `swa-oBn` algorithm recovers the true DAG $\mathcal{G}^*$ with probability tending to one.*

REMARK 6 (Heterogeneous ordinal scales). *The separation of skeleton discovery and edge orientation has theoretical significance beyond implementation convenience. Under a single-score approach, within-equivalence-class discrimination requires the penalty terms to* *cancel; this holds exactly when all variables share a common number of levels L, and only approximately otherwise, with a remainder of order $O(\rho_n)$ that is asymptotically negligible relative to the $O(N)$ likelihood ratio but may affect finite-sample performance. The two-step procedure eliminates this issue entirely: orientation is governed by the unpenalized pseudo-likelihood (Theorem 5.2), so the number of ordinal levels plays no role in edge direction decisions.*

6. Simulation Studies. We evaluate the finite-sample performance of `swa-oBn` through two simulation studies. Study 1 assesses the impact of survey design on DAG recovery; Study 2 examines the value of covariate adjustment under confounding. A third study, demonstrating that the survey-weighting problem extends to other algorithm families (PC, FCI, LiNGAM), is presented in the Supplementary Material.

6.1. *Common Setup.* All simulations share the following design. A finite population of $N = 20{,}000$ units is generated from an ordinal Bayesian network with $q = 8$ nodes, each having $L = 3$ ordinal levels, matching the three-level response scale of the CAHPS social need items. The true DAG is drawn once from an Erdős–Rényi model with edge probability 0.25 and held fixed across replications within a study. Parent effects $\beta_{jk\ell}$ and intercepts α_j are drawn independently from $\mathrm{N}(0, \sigma^2)$, where σ controls signal strength; cutpoints are set to produce approximately balanced marginal distributions.

From the population, a probability sample of approximate size n is drawn via Bernoulli sampling with inclusion probabilities that depend on two designated oversampling variables. Units in the top quartile of their combined score receive elevated inclusion probabilities; remaining units receive lower probabilities. Three design tiers are considered: *None* (simple random sampling), *Mild* (approximately 4:1 inclusion-probability ratio between oversampled and remaining units), and *Strong* (12:1 ratio). Each scenario is replicated 100 times. Performance is measured by the structural Hamming distance (SHD) between the estimated and true DAGs, defined as the number of edge additions, deletions, or reversals required to transform one into the other (Ni, Chen and Wang, 2025). Edge precision and correct orientation rates are reported in the Supplementary Material.

6.2. *Study 1: Effect of Survey Design on Structure Recovery.* Study 1 compares `swa-oBn` with the naive oBN of Ni, Chen and Wang (2025) through three sub-studies, each varying one factor while holding the others fixed (Figure 1).

Panel (A) varies design strength at $n \approx 1{,}000$, $\sigma = 1.5$. Under simple random sampling, the two methods coincide (SHD ≈ 2.8), confirming that subsequent differences arise from the weight correction rather than implementation artifacts. As the design becomes more informative, the naive SHD rises monotonically to ≈ 4.5 under the Strong design, while `swa-oBn` remains at ≈ 3.5 — a reduction of roughly one structural error.

Panel (B) varies sample size under the Strong design. Both methods improve with n, but `swa-oBn` converges faster: at $n = 2{,}000$, SHD ≈ 3.3 versus ≈ 4.0 for the naive method. The persistent gap at large n reflects a systematic bias in the naive estimator rather than sampling variance, consistent with Theorems 5.1 and 5.2.

Panel (C) varies signal strength under the strong design. `swa-oBn` achieves lower SHD across all signal levels, with the largest improvement at mild signals ($\sigma \in [0.5, 1.0]$) where the ordinal identifiability mechanism retains enough signal to orient edges while design-induced bias is strong enough to mislead the unweighted method.

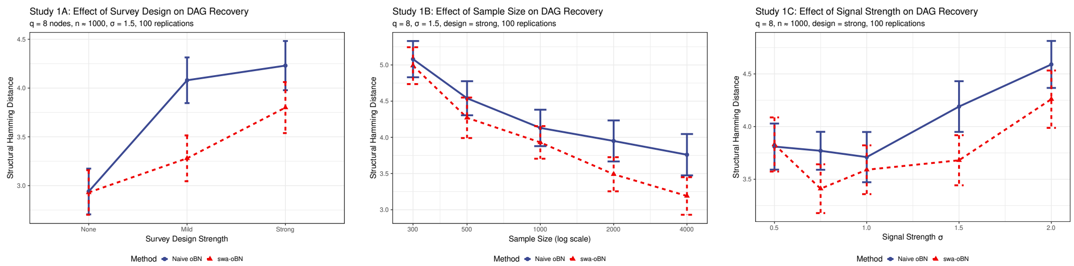

FIG 1. *Mean structural Hamming distance (± 1.96 SE) for* `swa-oBn` *(red, dashed) and the naive oBN of Ni, Chen and Wang (2025) (blue, solid) across 100 replications. (A) Varying survey design strength at $n \approx 1{,}000$, $\sigma = 1.5$; the two methods coincide under simple random sampling and diverge as the design becomes informative. (B) Varying sample size under the Strong design at $\sigma = 1.5$; the persistent gap as n grows indicates bias in the naive method. (C) Varying signal strength under the Strong design at $n \approx 1{,}000$. All panels: $q = 8$ and $L = 3$.*

6.3. *Study 2: Value of Covariate Adjustment.* Study 2 isolates the role of covariate adjustment and its interaction with survey weighting. The true DAG has $q = 8$ nodes partitioned into six upstream need and two terminal outcome nodes. Two parallel chains emanate from a common root ($X_1 \to X_2 \to X_3 \to X_4$ and $X_1 \to X_5 \to X_6$); outcome X_8 is a collider with parents X_4 and X_6 and outcome X_7 branches off X_3. A continuous confounder Z affects five of the eight nodes (three upstream needs and both outcomes); its effect magnitude is governed by δ. Sampling uses the Strong design with $\sigma = 1.5$ and $n \approx 1{,}000$. Four estimators are compared: (i) *Naive* (no weights, no $\boldsymbol{Z}$); (ii) $\boldsymbol{Z}$*-adjusted* (no weights, $\boldsymbol{Z}$ included); (iii) *Weighted* (weights, no $\boldsymbol{Z}$); and (iv) `swa-oBn` (weights and $\boldsymbol{Z}$).

Figure 2(A) varies confounding strength δ. At $\delta = 0$, all four methods perform similarly (SHD ≈ 6.3–6.9). As δ grows, the Naive and Weighted estimators degrade in parallel to ≈ 8.7 at $\delta = 2.0$, confirming that weighting alone does not address confounding. The $\boldsymbol{Z}$-adjusted estimator improves with stronger confounding (SHD decreasing to ≈ 5.7), as conditioning on $\boldsymbol{Z}$ removes spurious edges. The full `swa-oBn` is lowest at every level above $\delta = 0.5$, reaching ≈ 5.3 at $\delta = 2.0$. That $\boldsymbol{Z}$-adjustment offers no gain at $\delta = 0$ confirms its role is confounding-specific; the gain from weighting under informative designs without confounding is established in Study 1.

Figure 2(B) fixes strong confounding ($\delta = 1.2$) and varies the design. Under simple random sampling, covariate adjustment dominates: the $\boldsymbol{Z}$-adjusted and `swa-oBn` estimators both achieve SHD ≈ 4, versus ≈ 7 for the Naive and Weighted methods. As the design strengthens, the $\boldsymbol{Z}$-adjusted estimator degrades ($\approx 4.3 \to 6.0$) because it cannot correct design-induced bias, while `swa-oBn` maintains its advantage. Survey weighting and covariate adjustment thus address distinct bias sources and must be combined for reliable DAG recovery.

6.4. *Study 3: Generality of the Survey-Weighting Problem.* To confirm that the distortion introduced by informative sampling is not specific to ordinal Bayesian networks, we applied survey-weighted and unweighted versions of three additional algorithms — the PC algorithm, FCI, and LiNGAM — to a simple $X \to Y \to Z$ chain. Under Mild and Strong designs, the unweighted versions frequently infer a spurious X–Z edge, while their weighted counterparts recover the correct structure. Complete results appear in the Supplementary Material.

7. Application: Causal Architecture of Veteran Social Determinants of Health. We apply the `swa-oBn` to the VHA CAHPS data of Section 2 to address the scientific questions posed therein. Sensitivity analyses concerning the proportional odds working assumption are deferred to the Supplementary Material.

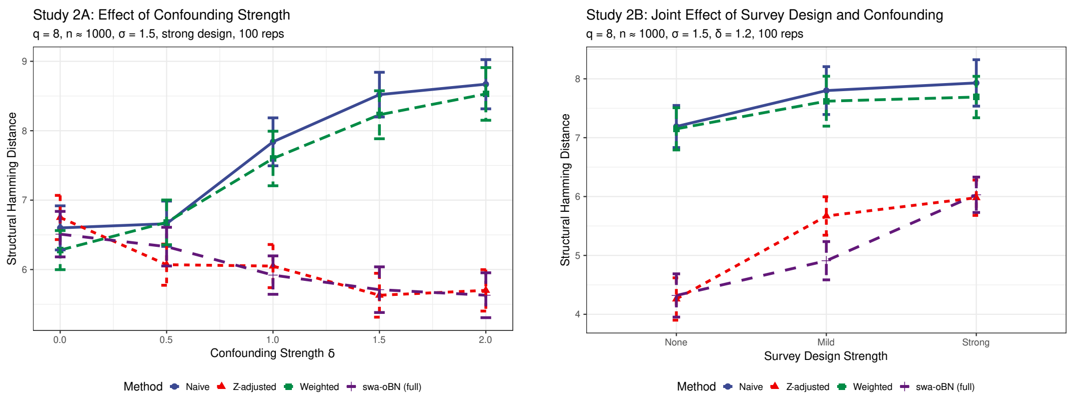

FIG 2. *Mean SHD (± 1.96 SE) for four estimators (Naive; $\boldsymbol{Z}$-adjusted; Weighted; full* `swa-oBn`*) across 100 replications. (A) Varying confounding strength δ under the Strong design; weighting alone fails to address confounding, while the full* `swa-oBn` *is lowest throughout. (B) Varying design strength with strong confounding fixed at $\delta = 1.2$; the $\boldsymbol{Z}$-adjusted estimator degrades as the design becomes informative, while* `swa-oBn` *holds steady. Both panels: $q = 8$ nodes, $\sigma = 1.5$, $n \approx 1{,}000$.*

7.1. *Point-estimate DAG and bootstrap uncertainty.* We ran `swa-oBn` with $R = 10$ random restarts, the covariates $\boldsymbol{Z}$ of Section 2.4, and the outcome-to-need blacklist. Kish's effective sample size is $n_{\text{eff}} = 1{,}387$ (design effect 3.89), so the pseudo-BIC penalty scales with $\log n_{\text{eff}} \approx 7.23$. Uncertainty was quantified by $B = 250$ nonparametric pairs bootstrap resamples (Section 4.4).

Figure 3 displays the bootstrap median-probability-model (MPM) DAG (Barbieri and Berger, 2004), retaining directed edges with inclusion probability $\hat{p}_{jk} \geq 0.50$ and classifying them as *robust* ($\hat{p}_{jk} \geq 0.75$) or *suggestive* ($0.50 \leq \hat{p}_{jk} < 0.75$). Table 1 reports the 14 consensus edges (8 robust, 6 suggestive); the full 10×10 matrix appears in Supplementary Table S2. Nine of ten nodes are connected; *childcare* is isolated, reflecting low prevalence of childcare needs in an older veteran sample. The edge between mental health and well-being is present in every resample but its orientation is not identified ($\hat{p}_{\text{WB}\to\text{MH}} = 0.54$ vs. $\hat{p}_{\text{MH}\to\text{WB}} = 0.46$), as expected for cross-sectional data on simultaneously-determined outcomes.

7.2. *Weighted versus unweighted comparison.* To assess the impact of survey weighting, we re-ran the analysis with all weights set to one, holding the covariate adjustment, blacklist, restarts, and bootstrap resamples fixed. The unweighted analysis retains 18 consensus edges at $\hat{p}_{jk} \geq 0.50$ versus 14 for the weighted analysis: 11 edges are shared, 7 appear only unweighted, 3 appear only weighted, and one edge reverses direction. A side-by-side visualization and complete edge-level breakdown appear in Supplementary Section F.

The consequential difference concerns discrimination/isolation. In the weighted DAG it has out-degree 4 and sits as a proximal mediator downstream of basic needs; in the unweighted DAG its out-degree rises to 8, placing it as the root of the entire network with apparent causal influence on transportation, internet, legal assistance, and basic needs. Non-White-male veterans were over-sampled by the SHEP design and have elevated discrimination-related needs across most domains (Frank et al., 2025; Russell et al., 2026), so unweighted estimation reads the resulting sample-level correlations as a common cause; weighting back to the superpopulation reveals that discrimination/isolation is structurally a mediator. An investigator running the unweighted analysis would therefore conclude that material deprivation acts directly on well-being rather than through discrimination/isolation, a qualitatively different policy conclusion. The pattern is consistent with the simulation evidence of Section 6.2.

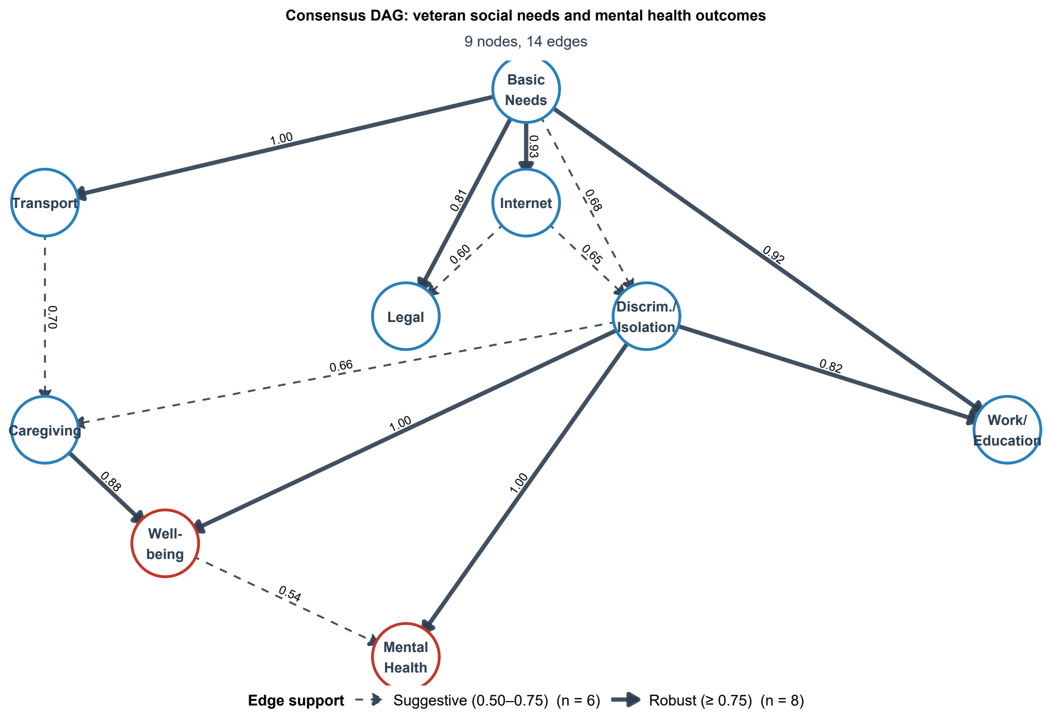

FIG 3. *Consensus DAG for veteran social needs and mental health outcomes, estimated by* `swa-oBn` *on the VA SHEP January–February 2023 primary care sample ($n = 5{,}396$), adjusted for age and race-ethnicity-sex and weighted by SHEP sampling weights. Outcome-to-need edges forbidden by blacklist. Edge probabilities are bootstrap inclusion frequencies over $B = 250$ resamples (all converged). Displayed edges have $\hat{p}_{jk} \geq 0.50$; dashed edges are* suggestive *($\hat{p} \in [0.50, 0.75)$), solid are* robust *($\hat{p} \geq 0.75$).*

7.3. *Substantive findings.* The weighted consensus DAG yields three findings addressing the scientific questions of Section 2.

Unmet basic needs are the upstream hub. Basic needs has direct edges to five of the seven other need nodes to which it can connect, with inclusion probabilities from 0.68 to 1.00; no other need variable has comparable out-degree. Basic financial deprivation is therefore the root cause in the social-need network, and interventions addressing food, housing, and basic necessities have the greatest potential for cascading downstream benefits. This interpretation is conditional on causal sufficiency; unmeasured confounders such as income or geography could alter the estimated structure.

Discrimination and isolation are the proximal mediators of health. Edges from discrimination/isolation to well-being and mental health both carry $\hat{p} = 1.00$, the highest in the graph; most individual needs have no direct edge to either outcome. Material deprivation thus reaches health primarily through discrimination and isolation, so interventions targeting material needs alone may fail to improve outcomes. As shown in Section 7.2, this finding is conditional on appropriate survey weighting.

Mental health is reached through mediation, not directly. Mental health receives no direct edges from individual needs; the primary pathway is needs $\to$ discrimination/isolation $\to$ mental health, with a secondary contribution via caregiving $\to$ well-being $\to$ mental health. The caregiving $\to$ well-being edge ($\hat{p} = 0.88$) is noteworthy because caregiving burden is not among the nine social risk domains assessed by the VHA's ACORN screening initiative (Russell et al., 2023) or the five core domains of the CMS Accountable Health Communities

TABLE 1
Bootstrap inclusion probabilities for the 14 consensus edges of Figure 3 ($B = 250$). Robust: $\hat{p}_{jk} \geq 0.75$; suggestive: $\hat{p}_{jk} \in [0.50, 0.75)$. Full 10×10 matrix in Supplementary Table S2.

From	To	$\hat{p}_{jk}$	Tier
Discrimination/Isolation	Mental Health	1.00	Robust
Basic Needs	Transport	1.00	Robust
Discrimination/Isolation	Well-being	1.00	Robust
Basic Needs	Internet	0.93	Robust
Basic Needs	Work/Education	0.92	Robust
Caregiving	Well-being	0.88	Robust
Discrimination/Isolation	Work/Education	0.82	Robust
Basic Needs	Legal	0.81	Robust
Transport	Caregiving	0.70	Suggestive
Basic Needs	Discrim./Isolation	0.68	Suggestive
Discrimination/Isolation	Caregiving	0.66	Suggestive
Internet	Discrim./Isolation	0.65	Suggestive
Internet	Legal	0.60	Suggestive
Well-being	Mental Health	0.54	Suggestive

HRSN tool (Billioux et al., 2017), yet the DAG identifies it as a proximal determinant of well-being.

8. Discussion. This paper introduced the survey-weighted, covariate-adjusted ordinal Bayesian network (`swa-oBn`), a framework for causal discovery from complex survey data with ordinal responses. The method integrates Horvitz–Thompson weights and Kish's effective sample size into a pseudo-BIC scoring framework, develops a two-step scoring procedure that separates skeleton discovery from edge orientation, supports exogenous covariate adjustment without expanding the DAG, and allows domain knowledge to be encoded via structural constraints. We proved that the framework consistently recovers the true population-level causal graph under a joint superpopulation-and-design asymptotic framework, with skeleton consistency established via the penalized pseudo-BIC and orientation consistency via the unpenalized pseudo-likelihood.

Applied to the VHA CAHPS survey, the `swa-oBn` revealed a causal architecture of Veteran social determinants of health that could not have been discovered by the associational methods used in prior analyses of this dataset (Frank et al., 2025; Lamba et al., 2025; Russell et al., 2026). Three principal findings emerged. First, unmet basic needs (food, housing, paying for basics) function as the upstream causal hub, with direct edges to five of the seven other need nodes. Second, discrimination and social isolation serve as the proximal mediator of health outcomes, with bootstrap inclusion probabilities of 1.00 for both the well-being and mental health edges. Third, mental health is reached through mediation, primarily via discrimination/isolation and degraded well-being, rather than directly from most individual needs. These findings provide the directional knowledge required for intervention prioritization: addressing basic needs has the greatest potential for cascading downstream benefits, but interventions that do not also address discrimination and isolation may fail to reach health outcomes through the primary mediating channel.

The demographic covariates $\boldsymbol{Z}$ (age and the combined race-ethnicity-sex indicator) are the same variables adjusted for in all three companion analyses of this dataset, where they substantially altered prevalence ratios (Frank et al., 2025) and risk-need concordance patterns (Russell et al., 2026). Age captures cohort and life-stage effects; the race-ethnicity-sex composite absorbs the systematic disparities documented in those studies and reflects the stratified oversampling design. These are plausibly the dominant demographic confounders for the relationships among social need variables in this population. However, several variables that could confound need-need or need-outcome relationships remain unmeasured in the CAHPS data. We do not claim that causal sufficiency holds perfectly. Rather, we argue that the $\boldsymbol{Z}$-adjusted DAG is more credible than the unadjusted one and that the methodology is correct conditional on the assumption. If richer covariate data become available in future CAHPS administrations, they can be incorporated into the `swa-oBn` framework with no algorithmic changes; only the $\boldsymbol{Z}$ matrix needs to be expanded.

Several additional limitations merit discussion. First, the greedy hill-climbing search is guaranteed to find only a local optimum. The multi-start procedure with $R = 10$ random initializations mitigates this concern, and the bootstrap procedure implicitly explores the DAG space by re-running the search on resampled data; nevertheless, the estimated DAG is not guaranteed to be the global optimum. Second, the polychoric clustering used to construct composite nodes introduces a working-model approximation. The pointwise maximum across constituent three-level items, while still a three-level ordinal, does not in general satisfy an exact cumulative link specification conditional on its parents, so the identifiability guarantee of Ni and Mallick (2022) holds only approximately for these composite nodes. We treat this as a deliberate trade-off: the alternative (entering all 15 raw items into the graph) produces an overly dense and substantively uninterpretable structure. The bootstrap procedure provides some robustness against orientation artifacts arising from this approximation. Third, the global effective sample size n_{eff} used in the pseudo-BIC penalty does not capture node-specific design effects. Different variables may have different design effects depending on how strongly the oversampling correlates with each variable. Incorporating node-specific design effects would require estimating variable-level design effects and integrating them into the scoring, an extension we leave for future work. Fourth, the naive nonparametric bootstrap does not exactly replicate the variance structure of the stratified sampling design; a design-consistent alternative such as the rescaled bootstrap of Rao and Wu (1988) would improve the calibration of intermediate-valued edge inclusion probabilities. Fifth, the CAHPS data are cross-sectional, so the discovered DAG represents a contemporaneous causal structure, not a temporal one. The blacklist encoding the assumption that outcomes do not cause needs is the minimal temporal constraint imposed; all other directionality is learned from the data. Sixth, the proportional odds structure of the cumulative link model is treated as an explicitly designated working assumption rather than a tested property of the data; the Supplementary Material documents diagnostic checks and the anti-conservatism of standard diagnostics under complex survey designs.

The survey-weighting problem documented in this paper is not specific to ordinal Bayesian networks. The Supplementary Material demonstrates that the PC algorithm, FCI, and LiNGAM all suffer from analogous distortions when applied to complex survey data. Developing fully survey-weighted extensions of these algorithms for multivariate settings (with formal consistency guarantees comparable to those established here for the `swa-oBn`) is a natural next step and would provide practitioners with a complete toolkit for design-consistent causal discovery. A further methodological priority is sensitivity analysis for unmeasured confounding in causal discovery; adapting E-value methods (VanderWeele and Ding, 2017) from treatment effect estimation to graph-valued target would provide a principled framework for assessing robustness of the learned DAG. The two-step scoring procedure, which cleanly separates skeleton control from orientation control, may also be of independent methodological interest for ordinal BN learning beyond the survey-weighted setting and could be developed as a standalone contribution. Finally, longitudinal extensions of the CAHPS survey, if available, would enable temporal causal discovery and permit relaxation of the cross-sectional assumption that currently limits the interpretation of the discovered graph.

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SUPPLEMENTARY MATERIAL

Supplementary Material for "Causal Discovery from Complex Survey Data"
Full proofs of Propositions 1–2 and Theorems 1–2; polychoric clustering details; Study 3 simulation results (survey-weighted PC, FCI, LiNGAM); complete bootstrap edge probability matrix; sensitivity analyses; weight distribution details.

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